Unit 3: Measurement — Challenge Pack

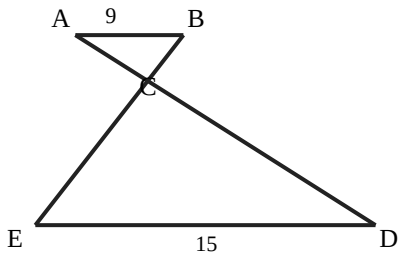
Math 9E · Similar Triangles · Unit Conversions · Surface Area · Volume & Capacity · Scale & Isometric Drawings · Trigonometry review

How to use: Part A has very hard questions for each lesson. Part B has the challenging questions your teacher is most likely to ask, based on the patterns in your worksheets (the ★★★ and ★★★★ problems, the "Challenge before next class" and the proofs). Worked answers are in Part C at the end. ★★★ = hard, ★★★★ = very hard.

PART A — Very Hard Questions by Lesson

A1. Similar Triangles & Thales' Theorem (Lesson 3.1)

A1.1 ★★★ Segments AD and BE cross at C (an "hourglass"). AB ∥ ED, AB = 9 cm, ED = 15 cm and AD = 20 cm. Find AC and CD.



A1.2 ★★★ In ΔABC, DE ∥ BC with D on AB and E on AC. AD = x, DB = x + 3, AE = 4 and EC = 6. (a) Find x. (b) If DE = 8, find BC.

A1.3 ★★★ Two similar triangles have perimeters 24 cm and 40 cm. The larger triangle has an area of 75 cm². Find the area of the smaller triangle.

A1.4 ★★★★ ΔABC has a right angle at C. CD is the altitude to the hypotenuse, AB = 25 and CD = 12. Find AD and DB (AD < DB), then the legs AC and BC, the perimeter and the area of ΔABC.

A1.5 ★★★★ A 1.6 m student stands 6 m from a lamp post and casts a 2 m shadow. (a) How tall is the lamp post? (b) The student walks 2 m farther away from the post. How long is the new shadow?

A1.6 ★★★★ A conical tank (point down) is 12 ft deep and the diameter of its top is 8 ft. (a) To what depth must it be filled so that it holds exactly **half** its capacity? (b) What is the area of the water surface at that depth?

A1.7 ★★★★ **Proof.** In the hourglass figure of A1.1 (AB ∥ ED), prove that ΔABC ∼ ΔDEC, and then prove that AC · CE = BC · CD.

A2. Unit Conversions (Lesson 3.2)

A2.1 ★★★ A farm has an area of 2.5 km². Convert this to (a) m², (b) hectares (1 ha = 10 000 m²), (c) acres.

A2.2 ★★★ Convert 7.2×10^{-4} km to µm, and 0.003 Gm to mm. Write both answers in scientific notation.

A2.3 ★★★ A car travels at 90 km/h. Convert this to (a) m/s, (b) miles per hour.

A2.4 ★★★★ 1 ft = 30.48 cm. (a) How many cm^2 are in 1 ft^2? (b) A room is 12 ft × 15 ft. Find its area in m^2, to the nearest hundredth.

A2.5 ★★★★ A doctor orders 15 mg of medicine per kg of body mass for a child with a mass of 44 lb (1 kg = 2.2 lb). The syrup label reads 250 mg in 5 mL. How many mL should the nurse give?

A2.6 ★★★★ Convert 3.2 m^3 to cm^3 and to litres. Explain why 1 $m^3 \neq 100\ cm^3$.

A3. Surface Area (Lesson 3.2)

A3.1 ★★★ A cone has radius 6 cm and **height** 8 cm (not slant height). Find its total surface area in terms of π and to the nearest tenth.

A3.2 ★★★ A closed cylinder has radius 5 cm and total surface area 120π cm^2. Find its height.

A3.3 ★★★★ A cone has a curved surface area of 150π cm^2 and a slant height of 15 cm. Find the radius, then the height to the nearest hundredth.

A3.4 ★★★★ A grain silo is a cylinder (radius 3 m, height 10 m) topped by a hemisphere. Only the outside walls and the dome are painted (not the base). One can of paint covers 12 m^2 and costs $38. Find the area painted, the number of cans and the total cost with 12% tax.

A3.5 ★★★ A right square pyramid has base side 10 m and height 12 m. Find its slant height and total surface area.

A3.6 ★★★★ A pipe is 20 cm long with outer radius 5 cm and inner radius 3 cm. Find the total surface area: outside, inside and both ring-shaped ends.

A3.7 ★★★ A solid hemisphere has total surface area 75π cm^2. Find its radius. (Remember the flat face!)

A4. Volume & Capacity (Lesson 3.3)

A4.1 ★★★★ A square pyramid has base 12 m × 12 m and height 18 m. It is cut horizontally 6 m from the apex and the top is removed. Find the volume of the remaining piece (frustum).

A4.2 ★★★ A sphere fits exactly inside a cube of side 10 cm. Find the volume of empty space and the percentage of the cube that is empty.

A4.3 ★★★★ A cone has volume 100π cm^3 and height 12 cm. Find its radius, slant height and total surface area.

A4.4 ★★★★ A metal sphere of radius 6 cm is melted and recast into small cones with radius 3 cm and height 4 cm. How many cones can be made?

A4.5 ★★★ A cylindrical tank has diameter 2.4 m and height 3 m. (a) Find its capacity in kL. (b) A pump fills it at 40 L/min. How many hours will it take?

A4.6 ★★★★ A pool is 25 m long and 10 m wide. Its depth slopes evenly from 1 m at one end to 3 m at the other end. Find its capacity in litres. How many days does it take to fill with a hose delivering 20 L/min?

A4.7 ★★★ An ice-cream cone has slant height 5 cm and radius 3 cm, topped by a hemisphere of ice cream of radius 3 cm. Find the total volume.

A4.8 ★★★★ A cube of side 10 cm has a cylindrical hole of radius 2 cm drilled straight through it, from the centre of one face to the centre of the opposite face. Find the remaining volume and the new total surface area.

A5. Scale, Scale Diagrams & Isometric Drawings (Lesson 3.4)

A5.1 ★★★★ A map has scale 1:25 000. (a) Two towns are 18 cm apart on the map. How far apart are they in km? (b) A park covers 12 cm² on the map. Find its actual area in m² and in hectares. (Careful: area does not use the same factor!)

A5.2 ★★★★ A model car is built at a scale of 1:18. The model's fuel tank holds 9 mL. How many litres does the real tank hold?

A5.3 ★★★ A real wall is 4.2 m long and is drawn as 3.5 cm on a plan. Write the scale as 1 : n. Is it a reduction or an enlargement?

A5.4 ★★★ A cell 6 µm wide appears 2.4 cm wide in a textbook image. Find the scale and the scale factor.

A5.5 ★★★★ A floor plan has scale 1:50. A room measures 9 cm × 7 cm on the plan. Flooring costs $42/m². Find the total cost with 12% tax.

A5.6 ★★★★ A solid is built from 1-cm cubes. The numbers on its top-view grid show how many cubes are stacked on each square (the back row is at the top).

3	**2**	**1**
2	**1**	
1		

(a) How many cubes are there? (b) Describe or draw the front view and the right view. (c) Find the total surface area of the solid.

A6. Trigonometry & Inscribed Prisms (review of earlier problems)

A6.1 ★★★★ The largest possible equilateral triangular prism is cut from a cylinder of radius 4 cm and length 10 cm. Use trigonometry to find the side of the triangle, then find the wasted volume and the percentage wasted.

A6.2 ★★★★ The largest square prism is cut from a cylinder of radius 5 cm and length 8 cm. Find the side of the square and the wasted volume.

A6.3 ★★★★ Compare the percentage of the cylinder that is used when the inscribed prism has a triangular, a square or a regular hexagonal cross-section. Which wastes the least? Explain why.

PART B — Predicted Challenging Test Questions

These questions combine skills the same way your worksheets do. The yellow note explains why each is likely.

B1 An underground tank is a right cone with its apex pointing down. Base diameter 5.0 m, height 3.5 m. (a) Find its capacity in kL to the nearest tenth. (b) How much water is in it when the water level is 1 m below the top? (c) What percentage full is it?

Why: Practice 3.3 #4 is a ★★★+ question. Teachers often reuse it with the "1 m below" twist, which needs similar cones (k^3).

B2 A right rectangular pyramid has base 5 m × 3 m and height 10 m. A horizontal cut is made 4 m from the apex and the top pyramid is removed. Find the volume of the remaining piece.

Why: Practice 3.3 #9 (★★★+) with different numbers.

B3 A silo is a cylinder with a conical roof. Diameter 10 m, cylinder height 15 m, roof height 4 m. The walls and roof need sealant (not the base). One bucket covers 30 m^2 and costs $65. Find the area (2 decimal places), number of buckets and total cost.

Why: Practice 3.2 #8. You must find the slant height with Pythagoras first, which is where most students lose marks.

B4 A fish tank is 120 cm × 50 cm × 45 cm. Fancy goldfish need at least 75 US gallons (1 gal = 3.785 L). (a) Is the tank big enough? (b) Keeping length and width the same, what height would hold exactly 75 gallons?

Why: Practice 3.3 #6, extended with a reverse step.

B5 A doctor orders 0.6 g of syrup medicine. The label reads 250 mg in 5 mL. How many mL should be given? How many teaspoons is that (1 tsp = 5 mL)?

Why: Practice 3.2 #3–#4. The trap is the g → mg conversion.

B6 Two similar cones have volumes 54π cm^3 and 250π cm^3. (a) Find the ratio of their radii. (b) The small cone has surface area 36π cm^2. Find the surface area of the large cone.

Why: Practice 3.1 #7 (★★★+) used k^2 for areas. A test question may go one step further to k^3.

B7 A bathroom is drawn on grid paper where 4 cm represents 1 m. On the plan it is 16 cm × 10 cm. The walls are 3 m tall and have one door of 0.9 m × 2.1 m that is not painted. Tiling costs $225/$m^2$ and painting costs $16/$m^2$. Find the total cost including 12% tax.

Why: Practice 3.4 #8 (★★★★). It combines scale, area, perimeter and cost.

B8 Derivation. Use Cavalieri's principle with a hemisphere and a cylinder with a cone removed (same radius r and height r) to derive $V_{sphere} = (4/3)\pi r^3$.

Why: Practice 3.3 #13 (★★★★). Proofs like this are very common as the "challenge" question on a test.

B9 Derivation. Explain why the volume of a pyramid is one-third of the volume of a prism with the same base and height.

Why: it was the "Challenge before next class" on your answer-key page.

B10 In ΔEFD, points A and B lie on EF with EA = 4, AB = 10 and BF = 6. Point C is inside the triangle with AC $\parallel$ ED and BC $\parallel$ FD. FD = 16. (a) Prove $\Delta ABC \sim \Delta EFD$. (b) Find BC.

Why: Practice 3.1 #9 asks for a similarity proof plus finding x.

B11 A rectangular lawn is 30 m × 20 m. A mower cuts a 50-cm strip and mows back and forth along the 30 m length. Find the distance travelled in km and the time in minutes at 4 km/h.

Why: Practice 3.2 #9 (★★★★), a simpler shape with the same reasoning.

B12 In right ΔABC (right angle at C), CD $\perp$ AB. Prove $CD^2 = AD \cdot DB$. Then, if AD = 3.6 and DB = 6.4, find CD, AC and BC.

Why: you were asked to prove this in class. Expect a proof followed by numbers.

PART C — Answers & Worked Solutions

A1. Similar Triangles

A1.1 AB ∥ ED ⇒ ΔABC ~ ΔDEC (AA: vertical angles at C, alternate angles). AC/CD = 9/15 = 3/5. AC = 20 · 3/8 = **7.5 cm**, CD = **12.5 cm**.

A1.2 (a) x/(x+3) = 4/6 ⇒ 6x = 4x + 12 ⇒ **x = 6**. (b) AB = 15. DE/BC = AD/AB = 6/15 ⇒ BC = 8 · 15/6 = **20**.

A1.3 k = 24/40 = 0.6, so the area ratio is $k^2 = 0.36$. Small area = 75 × 0.36 = **27 cm²**.

A1.4 AD · DB = 144 and AD + DB = 25 ⇒ $t^2 - 25t + 144 = 0$ ⇒ **AD = 9, DB = 16**. $AC^2 = 9 \cdot 25 = 225$ ⇒ **AC = 15**. $BC^2 = 16 \cdot 25 = 400$ ⇒ **BC = 20**. Perimeter = **60**. Area = ½ · 15 · 20 = **150** (check: ½ · 25 · 12 = 150 ✓).

A1.5 (a) The shadow tip is 6 + 2 = 8 m from the post. H/8 = 1.6/2 ⇒ **H = 6.4 m**. (b) Now the student is 8 m away: 1.6/s = 6.4/(8 + s) ⇒ 6.4s = 12.8 + 1.6s ⇒ 4.8s = 12.8 ⇒ **s ≈ 2.67 m**.

A1.6 (a) The water forms a cone similar to the tank. Volume ratio $k^3 = 1/2$ ⇒ $k = 1/\sqrt[3]{2} \approx 0.7937$. Depth = 12 · 0.7937 ≈ **9.52 ft** (not 6 ft!). (b) r = 4 · 0.7937 ≈ 3.175 ft. Area = $\pi r^2 \approx$ **31.67 ft²**.

A1.7 ∠ACB = ∠DCE (vertical angles). ∠BAC = ∠EDC (alternate angles, AB ∥ ED). So ΔABC ~ ΔDEC (AA). Corresponding sides: AC/DC = BC/EC. Cross-multiply: **AC · EC = BC · DC** ∎

A2. Unit Conversions

A2.1 1 km² = (1000 m)² = 10^6 m². (a) **2 500 000 m²** (b) ÷ 10 000 = **250 ha** (c) × 2.47 = **617.5 acres**.

A2.2 7.2×10^{-4} km = 0.72 m = **7.2×10^5 µm**. 0.003 Gm = 3×10^6 m = **3×10^9 mm**.

A2.3 (a) 90 × 1000 ÷ 3600 = **25 m/s** (b) 90 ÷ 1.609 ≈ **55.9 mph**.

A2.4 (a) $30.48^2 \approx$ **929.03 cm²** (b) 180 ft² × 0.3048^2 m² ≈ **16.72 m²**.

A2.5 44 ÷ 2.2 = 20 kg. Dose = 15 × 20 = 300 mg. Volume = 300 × 5/250 = **6 mL**.

A2.6 1 m³ = 100 cm × 100 cm × 100 cm = 10^6 cm³. 3.2 m³ = **3 200 000 cm³ = 3200 L**. You must cube the factor because there are three dimensions.

A3. Surface Area

A3.1 $s = \sqrt{6^2 + 8^2} = 10$. SA = π(6)(10) + π($6^2$) = **96π ≈ 301.6 cm²**.

A3.2 2π(25) + 2π(5)h = 120π ⇒ 50 + 10h = 120 ⇒ **h = 7 cm**.

A3.3 πr(15) = 150π ⇒ **r = 10 cm**. $h = \sqrt{15^2 - 10^2} = \sqrt{125} \approx$ **11.18 cm**.

A3.4 Walls 2π(3)(10) = 60π. Dome = ½ · 4π(3^2) = 18π. Total = 78π ≈ **245.04 m²**. 245.04 ÷ 12 = 20.4 ⇒ **21 cans** (always round up). 21 × 38 = $798; with tax ×1.12 = **$893.76**.

A3.5 Slant = $\sqrt{12^2 + 5^2}$ = **13 m**. SA = 10^2 + 4 · ½ · 10 · 13 = 100 + 260 = **360 m²**.

A3.6 Outside 2π(5)(20) = 200π; inside 2π(3)(20) = 120π; ends 2 · π($5^2 - 3^2$) = 32π. Total = **352π ≈ 1105.8 cm²**.

A3.7 Curved part $2\pi r^2$ + flat circle $\pi r^2 = 3\pi r^2 = 75\pi$ ⇒ $r^2 = 25$ ⇒ **r = 5 cm**.

A4. Volume & Capacity

A4.1 $V_{big} = \frac{1}{3} \cdot 144 \cdot 18 = 864$ m³. $k = 6/18 = 1/3 \Rightarrow V_{small} = 864/27 = 32$ m³ (check: base 4 × 4, $\frac{1}{3} \cdot 16 \cdot 6 = 32$ ✓). Remaining = **832 m³**.

A4.2 Sphere r = 5: $V = (4/3)\pi(125) \approx 523.60$. Empty = 1000 − 523.60 ≈ **476.40 cm³ ≈ 47.6%**.

A4.3 $\frac{1}{3}\pi r^2(12) = 100\pi \Rightarrow r^2 = 25 \Rightarrow$ **r = 5**. $s = \sqrt{(25 + 144)} =$ **13**. $SA = \pi(5)(13) + 25\pi =$ **90π ≈ 282.7 cm²**.

A4.4 $V_{sphere} = (4/3)\pi(216) = 288\pi$. $V_{cone} = \frac{1}{3}\pi(9)(4) = 12\pi$. $288\pi \div 12\pi =$ **24 cones**.

A4.5 (a) $V = \pi(1.2^2)(3) \approx 13.57$ m³ ≈ **13.6 kL**. (b) 13 572 L ÷ 40 ≈ 339.3 min ≈ **5.65 h**.

A4.6 The side face is a trapezoid: $(1 + 3)/2 \times 25 = 50$ m². $V = 50 \times 10 = 500$ m³ = **500 000 L**. Time = 25 000 min ≈ **17.4 days**.

A4.7 $h = \sqrt{(5^2 - 3^2)} = 4$. Cone $\frac{1}{3}\pi(9)(4) = 12\pi$; hemisphere $(2/3)\pi(27) = 18\pi$. Total = **30π ≈ 94.25 cm³**.

A4.8 $V = 1000 - \pi(2^2)(10) = 1000 - 40\pi \approx$ **874.34 cm³**. $SA = 600 - 2(4\pi) + 2\pi(2)(10) = 600 + 32\pi \approx$ **700.53 cm²**. The SA *increases* because the inside wall of the hole is added.

A5. Scale & Isometric

A5.1 (a) 18 × 25 000 = 450 000 cm = **4.5 km**. (b) Area factor = $25\,000^2 = 6.25 \times 10^8$. 12 cm² × $6.25 \times 10^8 = 7.5 \times 10^9$ cm² = **750 000 m² = 75 ha**.

A5.2 Volume factor = $18^3 = 5832$. 9 mL × 5832 = 52 488 mL ≈ **52.5 L**.

A5.3 3.5 cm : 420 cm = **1 : 120**, a **reduction**.

A5.4 2.4 cm = 24 000 µm. 24 000 : 6 = **4000 : 1** (enlargement). Scale factor = **4000**.

A5.5 Room 4.5 m × 3.5 m = 15.75 m². 15.75 × 42 = \$661.50; × 1.12 = **\$740.88**.

A5.6 (a) 3+2+1+2+1+1 = **10 cubes**. (b) Front view: columns of heights 3, 2, 1 from left to right (a staircase). Right view: heights 1, 2, 3 from left to right (front row on the left). Top view: an L-shaped staircase of 6 squares. (c) Top + bottom = 2 × 6 = 12; front + back = 2 × 6 = 12; left + right = 2 × 6 = 12 ⇒ **36 cm²**.

A6. Trigonometry & Inscribed Prisms

A6.1 From $\cos 30° = (a/2)/R$: $a = 2R \cos 30° = R\sqrt{3} =$ **4√3 ≈ 6.93 cm**. $V_{cyl} = 160\pi \approx 502.65$. $V_{prism} = (\sqrt{3}/4)(48)(10) = 120\sqrt{3} \approx 207.85$. Waste ≈ **294.81 cm³ ≈ 58.7%**.

A6.2 The diagonal of the square is the diameter 2R ⇒ side = $R\sqrt{2} =$ **5√2 ≈ 7.07 cm**. Square area = $2R^2 = 50$. Waste = $200\pi - 400 = 200(\pi - 2) \approx$ **228.32 cm³** (36.3%).

A6.3 Used: triangle $3\sqrt{3}/(4\pi) \approx$ **41.3%**; square $2/\pi \approx$ **63.7%**; hexagon $3\sqrt{3}/(2\pi) \approx$ **82.7%**. The hexagon wastes the least. The more sides the polygon has, the closer it fills the circle.

Part B — Predicted Questions

B1 (a) $V = \frac{1}{3}\pi(2.5^2)(3.5) \approx 22.91$ m³ ≈ **22.9 kL**. (b) Water depth = 2.5 m ⇒ $k = 2.5/3.5 = 5/7$. $V = 22.91 \times (5/7)^3 \approx$ **8.3 kL**. (c) $(5/7)^3 \approx$ **36.4%**. Being only 1 m below the top still leaves the tank well under half full!

B2 $V_{big} = \frac{1}{3}(15)(10) = 50$ m³. $k = 4/10 = 0.4 \Rightarrow V_{small} = 50 \times 0.064 = 3.2$ m³. Remaining = **46.8 m³**.

B3 r = 5. Roof slant = $\sqrt{(5^2 + 4^2)} = \sqrt{41} \approx 6.403$. Walls $2\pi(5)(15) = 150\pi$; roof $\pi(5)(6.403) \approx 32.02\pi$. Total ≈ **571.82 m²**. 571.82 ÷ 30 = 19.06 ⇒ **20 buckets ⇒ $1300**.

B4 (a) 120 × 50 × 45 = 270 000 cm³ = 270 L ÷ 3.785 ≈ 71.3 gal. **No, it is not big enough.** (b) 75 × 3.785 = 283.875 L = 283 875 cm³. h = 283 875 ÷ (120 × 50) ≈ **47.3 cm**.

B5 0.6 g = 600 mg. 600 × 5/250 = **12 mL = 2.4 tsp**.

B6 (a) $k^3 = 250/54 = 125/27$ ⇒ k = **5/3**, so radii are in the ratio 3 : 5. (b) Area ratio $k^2 = 25/9$ ⇒ $36\pi \times 25/9 =$ **100π cm²**.

B7 Room 4 m × 2.5 m. Floor = 10 m² × $225 = $2250. Walls: perimeter 13 m × 3 m = 39 m² − 1.89 m² (door) = 37.11 m² × $16 = $593.76. Subtotal $2843.76; with tax ×1.12 = **$3185.01**.

B8 Slice both shapes at height h. Hemisphere slice: radius $\sqrt{(r^2 - h^2)}$ (Pythagoras), so area = $\pi(r^2 - h^2)$. Washer: the outer radius is r. The cone (apex at the bottom, height r, radius r) has radius h at height h, so washer area = $\pi r^2 - \pi h^2 = \pi(r^2 - h^2)$. The areas are equal at every height, so by Cavalieri $V_{hemisphere} = V_{cyl} - V_{cone} = \pi r^3 - \frac{1}{3}\pi r^3 = \frac{2}{3}\pi r^3$. Sphere = $2 \times \frac{2}{3}\pi r^3 =$ **$(4/3)\pi r^3$** ∎

B9 A cube of side s can be split into **3 identical pyramids** that share one corner as their apex. Each has a square base s × s and height s, so $V = s^3/3 = \frac{1}{3}$(base area)(height). Stretching the cube into a box (or using Cavalieri) keeps the ratio 1/3, so $V_{pyramid} =$ **$\frac{1}{3}Bh$**.

B10 (a) AC ∥ ED ⇒ ∠CAB = ∠DEF (corresponding angles). BC ∥ FD ⇒ ∠CBA = ∠DFE. So ΔABC ∼ ΔEFD (AA). (b) EF = 4 + 10 + 6 = 20, k = AB/EF = 10/20 = 1/2 ⇒ BC = ½ · FD = **8**.

B11 Rows = 20 ÷ 0.5 = 40. Distance = 40 × 30 = 1200 m = **1.2 km**. Time = 1.2 ÷ 4 = 0.3 h = **18 min**.

B12 Proof: ∠A = ∠DCB (both equal 90° − ∠ACD) and both triangles have a right angle at D ⇒ ΔADC ∼ ΔCDB (AA) ⇒ AD/CD = CD/DB ⇒ $CD^2 = AD \cdot DB$ ∎. Numbers: $CD^2 = 3.6 \times 6.4 = 23.04$ ⇒ **CD = 4.8**. AB = 10. $AC^2 = 3.6 \times 10 = 36$ ⇒ **AC = 6**. $BC^2 = 6.4 \times 10 = 64$ ⇒ **BC = 8**.